

Q1. What will be the possible change in magnetic property (paramagnetic vs. diamagnetic) in B_2 and O_2 molecules if there were no sp mixing in B atomic orbitals and sp mixing in O atomic orbitals?

Q2. Draw bonding and antibonding MOs (σ and π) formed by the combination of s, $s(\sigma)$; p , $p(\sigma)$; and p , $p(\pi)$ atomic orbitals. Comment on g and u nature of the formed MOs.

Q3. What will happen (a) if F_2 gas is blown into water, and (b) if Ca is added into water?

Q4. There is a substance (MW = 400) in a solution (4 gm/l). The length of the cuvette is 2 cm and only 50% of the certain beam light is transmitted. What is the molar absorption co-efficient?

Q5. Calculate $\lambda_{\max}$ of the following compound (Given base value = 215 nm, substitution = 5 nm, exo = 5 nm, extra conjugation = 30 nm)

